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Review Article

Relationship among depression, fatigue, and sleep after traumatic brain injury: The role of physical exercise as a non-pharmacological therapy

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ABSTRACT

Traumatic brain injury (TBI) is a burdensome condition frequently associated with an increased risk of psychiatric disorders. Although the exact molecular signaling pathways have not yet been fully defined, the compromised integrity of functional brain networks in regions such as the prefrontal cortex and anterior cingulate cortex has been linked to persistent symptoms, including depression, fatigue, and sleep disorders. Understanding how TBI affects neural physiology enables the development of effective interventions. One such strategy may be physical exercise, which promotes neural repair and behavioral rehabilitation after TBI. However, there are caveats to consider when interpreting the effects of physical exercise on TBI-induced mental health issues. This review will highlight the main findings from the literature investigating how different physical exercise protocols affect the progression of TBI-induced depression, fatigue, and sleep disturbances. Furthermore, we aim to explore potential neurobiological pathways that explain how physical exercise influences depression, fatigue, and sleep following TBI.

1. Introduction

Traumatic brain injury (TBI) is recognized as a global public health issue, affecting 70 million people worldwide each year (Dewan et al., 2018). TBI is characterized by a combination of immediate mechanical dysfunction in brain tissue and secondary damage (Logsdon et al., 2015). This devastating condition is often accompanied by significant behavioral and cognitive impairments that can persist for an extended period (Juengst et al., 2024).

Epidemiological studies have reported that TBI is associated with an increased risk (up to 3.29 times) of new-onset psychiatric disorders compared to the general population (Alway et al., 2016; Mayer and Quinn, 2022). This multifactorial comorbidity is also linked to the severity and location of the neuronal injury (Shultz et al., 2017). While moderate and severe neuronal injuries lead to impulsivity, irritability, and apathy, mild TBI is associated with the worsening of symptoms of mental disorders such as post-traumatic stress disorder (PTSD), fatigue, and major depression (Brett et al., 2021; Englander et al., 2010a; Howlett et al., 2022a; Piantino et al., 2022). High prevalence of sleep disorders, excessive daytime sleepiness, and chronic fatigue have also been observed in patients with moderate to severe TBI (Schnieders et al., 2012).

Depression and PTSD symptoms may emerge from the acute to early chronic post-injury phases (Mayer and Quinn, 2022). However, it is not fully understood how neuronal injury influences brain architecture and neural metabolism across regions. Evidence from experimental models points to an interrelationship between cellular stress and structural abnormalities (Khatri et al., 2018b; Wright et al., 2017). These molecular

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associations following neuronal injury suggest the existence of a post-traumatic "window of vulnerability," during which behavioral deficits, such as depression-like behavior and sleep-wake disturbances, are observed in the chronic phase after TBI (da Silva Fiorin et al., 2024c; Morris et al., 2024; Walton et al., 2021a). A possible explanation for these altered behaviors is the acute failure of mitochondrial function, followed by increased inflammatory process and oxidative stress (Severo et al., 2020). This assumption is supported by correlations between reduced cerebral blood flow in the rostral anterior cingulate cortex and self-reported severe symptoms such as fatigue, anxiety, sleep disturbances, and impaired sleep in TBI patients, as analyzed 20 months after the injury (Thomas et al., 2021). In this context, early molecular alterations can lead to long-term morphological changes in the brain, disrupting normal neuronal neurotransmission and inducing changes in both functional and structural brain connectivity (da Silva Fiorin et al., 2024a).

In the sports environment, a relationship has been observed between long-term neurobehavioral sequelae, such as symptoms of mental distress (including depression and fatigue), and total or partial hypopituitarism (3,13). The growth hormone (GH) deficiency observed in experimental models of TBI also suggests that systemic inflammation and persistent astrocytosis in the hypothalamus and anterior pituitary may be possible causes of hypopituitarism (Englander et al., 2010b; Howlett et al., 2022b; Khatri et al., 2018; Walton et al., 2021b). However, the exact molecular signaling pathways have not yet been fully defined. In this context, it is crucial to understand how TBI affects neural physiology to design effective interventions that address the long-term burden of living with TBI.

One of these strategies may be supported by physical training, which, from a preclinical and prophylactic perspective, plays a role in reducing the onset of damage, limiting secondary neuronal death, and promoting neural repair and behavioral rehabilitation after TBI (Cassol et al., 2022; Papalia et al., 2022). Regarding the impact of physical exercise on brain health, cross-sectional studies have shown that regular physical exercise improves psychological health, enhances cognition, reduces the risk of developing depression, and helps prevent neurodegeneration (Dauwan et al., 2021; Pearce et al., 2022; Proschinger et al., 2022; Schuch et al., 2018). Although the neuronal modulation elicited by exercise provides a unique non-pharmacological therapeutic approach to control secondary damage (Mckee and Daneshvar, 2015; Royes and Gomez-Pinilla, 2019), limited information is currently available on the role of physical exercise in TBI-induced mental health issues. Therefore, this review will offer a concise overview of the molecular factors involved in the emergence of mental health conditions and how physical exercise influences the progression of these behavioral disorders after TBI.

2. Dynamic functional interface between pathophysiology and TBI related mental health symptoms

TBI is a neurological injury defined as an alteration in brain function or pathology caused by external force (Subramanian et al., 2022). This injury is often accompanied by significant behavioral disabilities and long-term medical complications, including permanent cognitive disorders, learning disabilities, and a broad range of behavioral and emotional issues (Royes and Gomez-Pinilla, 2019). Epidemiological studies suggest a 2–3-fold increase in the incidence of TBI-related disabilities. The pathophysiology is likely multifactorial, and it is increasingly viewed as a chronic disease process rather than a single event (Mckee and Daneshvar, 2015). However, without a broad and comprehensive understanding of the pathophysiological mechanisms leading to mental health conditions after TBI, developing targeted disease-modifying therapies remains challenging

It is important to note that the effectiveness of therapies can be influenced by an individual's self-awareness of the limitations associated with their injury. Clinical studies demonstrate that greater emotional dysfunction is associated with higher levels of awareness of the limitations caused by brain injury (Fleming et al., 1998; Godfrey et al., 1993) High levels of depression and anxiety are often observed in the chronic stages of TBI, driven by emotional distress resulting from self-awareness of daily limitations, including professional, social, and familial distancing (Dromer et al., 2021).

From a metabolic perspective, the neurological events caused by the primary injury often evolve into delayed secondary injuries, characterized by inflammation, mitochondrial dysfunction, excitotoxicity, and the apoptosis of neuronal and glial cells (Maas et al., 2008; Ng and Lee, 2019). Since inflammation persists for years after a TBI, it has been proposed that chronic low-grade inflammation is linked to the high rates of depression (ranging from 9 to 21 %) following a TBI (Bodnar et al., 2018; Bombardier et al., 2010; Coughlin et al., 2017; Stein et al., 2019).

Clinical studies have provided preliminary evidence linking TBI-induced inflammatory biomarkers in cerebrospinal fluid to an increased risk of developing depression (Juengst et al., 2015), as well as a higher likelihood of suicide attempts (Juengst et al., 2014; Madsen et al., 2018). Posttraumatic hypopituitarism observed in patients has also been associated with delayed recovery, impaired rehabilitation, and persistent neuropsychiatric symptoms, such as irritability and fatigue (Masel et al., 2017). Specifically, the cascade of hormonal deficits resulting from post-traumatic hypopituitarism can lead to debilitating fatigue, negatively impacting functional recovery in individuals with TBI (Kgosidialwa and Agha, 2019). Moreover, changes in hormones such as melatonin and cortisol, which are involved in sleep-wake homeostasis and circadian rhythms, can hinder rehabilitation and contribute to maladaptive lifestyle changes in TBI patients. It's important to note that the complex etiology of mental health manifestations in TBI—including a combination of brain dysfunction, psychological trauma, and the interplay between affective and physical symptoms—also compromises systemic physiology, affecting the functioning of various organs. This emerging phenomenon triggers a pathological loop that further impacts the brain and exacerbates the overall pathogenesis of TBI (Royes and Gomez-Pinilla, 2019).

Studies in experimental TBI models show that, along with hyperglycemia, a dysfunctional hypothalamic-pituitary-adrenal axis raises stress hormone levels (Tapp et al., 2019), promotes splanchnic hypoperfusion, and increases cytokine and chemokine levels following neuronal injury (Godinho et al., 2021). This systemic inflammatory process reduces immunological and bioenergetic functions, contributing to the persistence of syndromes such as depression, fatigue, and sleep disturbances (Sun et al., 2019). From this perspective, the present report proposes several molecular pathways that lead to psychiatric impairment after TBI in humans (Fig. 1). However, a more comprehensive understanding of the role of intracellular connections in neuropsychiatric symptoms after TBI remains to be elucidated.

3. The role of physical exercise on TBI-induced mental health symptoms

3.1. Molecular effects of physical exercise following TBI

It is important to note that injury-induced changes in metabolism affect synaptic plasticity (Royes and Gomez-Pinilla, 2019) and physical exercise increases energy demands (Cassilhas et al., 2016). Consequently, the severity of the injury may prolong the period of unresponsiveness to physiological activation, and early physical exercise may exacerbate TBI-induced consequences. A retrospective cohort study of student-athletes observed that high levels of cognitive and physical activity after injury negatively affected neurocognitive function and concussive symptoms (Majerske et al., 2008). These data support the idea that the probable range of intensity, duration, frequency, and type of exercise (aerobic, anaerobic, or physical therapy) after TBI must be considered to induce responses to important targets following neuronal injury. In this context, the implementation of physical exercise after TBI must be approached with caution. Acute neuronal reorganization

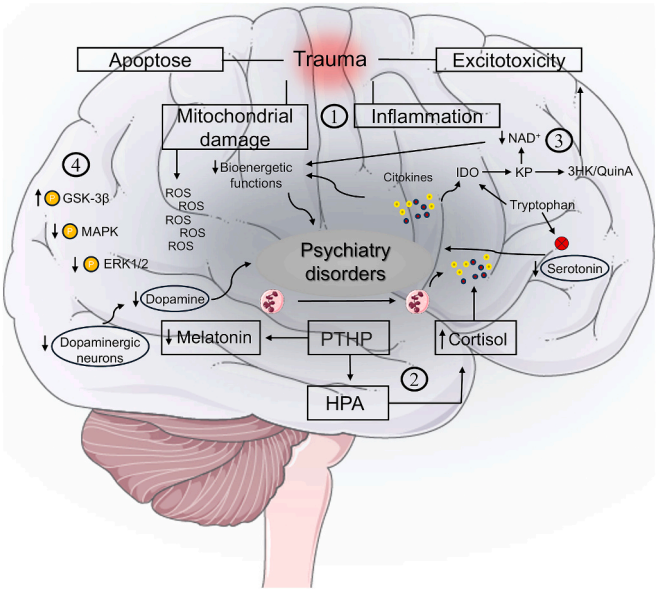

Fig. 1. The multifactorial pathophysiology of TBI in relation to psychiatric disorders following injury. Legend. 1) Several injuries, including inflammation, mitochondrial damage, excitotoxicity, and neuronal and glial cell apoptosis, are characteristic after TBI. 2) Post-traumatic hypopituitarism (PTHP) leads to a decrease in melatonin, which is involved in sleep-wake homeostasis and circadian rhythms. Furthermore, elevated stress hormone levels caused by the dysregulation of the hypothalamic-pituitary-adrenal (HPA) axis promote splanchnic hypoperfusion and increase cytokine and chemokine levels after neuronal injury. This amplified inflammatory process leads to reduced immunological and bioenergetic functions, compromising brain systems and contributing to the development of persistent syndromes such as depression, fatigue, and sleep disturbances. 3) The inflammatory process also increases indoleamine 2,3-dioxygenase (IDO) activity, promoting kynurenine pathway (KP) formation and decreasing nicotinamide adenine dinucleotide (NAD^+) levels, which results in bioenergetic dysfunction and increased reactive oxygen species (ROS) production. Concurrently, serotonin levels decrease, while the synthesis of 3-hydroxykynurenine (3HK) and quinolinic acid (QuinA) increases, leading to excitotoxicity. 4) Together, various molecular pathways—such as increased phosphorylation of the inhibitory site of glycogen synthase kinase-3 beta (GSK-3β), decreased levels of phosphorylated p38 mitogen-activated protein kinase (p-p38 MAPK) and phosphorylated extracellular signal-regulated kinase 1/2 (p-ERK1/2), the loss of dopaminergic neurons, and reduced dopamine levels—as well as systemic inflammation, complement the multifactorial process involved in psychiatric disorders following TBI.

following experimental TBI involves neurochemical and metabolic changes that engage several molecular pathways responsible for neuroplasticity (Griesbach et al., 2004). It is important to note that although TBI can induce early neuroplasticity, prolonged pathophysiological cascades are inevitable and persist over time, leading to cognitive deficits, as shown in experimental TBI models (da Silva Fiorin et al., 2016).

Moreover, neuroplasticity-related molecules, such as brain-derived neurotrophic factor (BDNF), synapsin I, and cAMP-response element-binding protein (CREB), decrease during the chronic period following injury, resulting in the loss of early repair mechanisms. It is possible to suggest that additional challenges to the brain following injury, such as physical exercise, can exacerbate neuronal impairment. For example, acute voluntary exercise in rats after TBI was shown to reduce levels of synapsin I, phospho-synapsin I, CREB, phospho-CREB, calcium/calmodulin-dependent protein kinase II (CAMKII), mitogen-activated protein kinase I and II, protein kinase C, and BDNF (Griesbach et al., 2004). Alongside these decreases in gene function and synaptic transmission molecules, greater cognitive deficits were observed compared to TBI rats that did not undergo acute exercise (Griesbach et al., 2004), suggesting an overload to the brain and potential ATP synthesis failure, leading to severe neuronal damage caused by early stimulation after TBI.

On the other hand, initiating exercise during a post-acute period (two weeks) after TBI has been shown to increase hippocampal levels of BDNF, CREB, and synapsin I in animal models, supporting the idea that physical exercise facilitates endogenous repair mechanisms and enhances functional recovery when applied within an appropriate time window (Griesbach, 2011). Indeed, plasticity-associated genes, including BDNF and CREB, are upregulated in the hippocampus during delayed exercise (five weeks) after experimental TBI, reducing the hippocampal inflammatory profile and increasing neuronal numbers (Piao et al., 2013).

However, it is important to note that differences in outcomes between acute and post-acute exercise after TBI may arise due to varying trauma severities and exercise intensities or types. For instance, progressive mild physical exercise initiated one week after a severe experimental injury reduces oxidative markers in the rat brain (Silva and Hoffmann, 2013), as well as mild acute physical activity within the first 14 days after a sport-related concussion has demonstrated beneficial effects in injured individuals (Leddy et al., 2023).

Since physical activity in individuals with TBI is a new area of research, examining appropriate measures and theories for understanding behavior should also be part of future studies. Given that physical exercise is a safe, non-pharmacological approach for mental health treatment, a better understanding of the molecular pathways induced by physical exercise can aid in developing more effective therapeutic strategies for treating these post-TBI behavioral changes.

4. The pathophysiology of post-TBI depression

After TBI, depression is one of the most common post-traumatic psychiatric disorders, with prevalence rates ranging from 6 % to 77 % (Robinson and Jorge, 2002). It has become a chronic public health issue (Hart et al., 2012; Kennedy et al., 2007). Epidemiological evidence reveals that TBI is associated with a significantly higher risk of developing depression, with nearly 11 % of individuals experiencing depression within one year following the injury (Choi et al., 2022). This rate is nearly eleven times higher than that of matched controls over the same period (Choi et al., 2022).

Depression treatment primarily involves pharmacological antidepressants and psychotherapies (Kendrick and Collinson, 2022). While these strategies are considered the frontline approaches (Kendrick and Collinson, 2022), response rates for most antidepressant medications range from 40 % to 60 % (Cipriani et al., 2018), and only about one-third of patients achieve remission with psychotherapy (Cuijpers et al., 2021).

The progressive loss of cortical gray matter in the frontal and temporal regions of TBI patients, even years after the injury (Bendlin et al., 2008; Hudak et al., 2011; Maller et al., 2014), supports the idea that compromised integrity of functional networks involved in emotional regulation contributes to the development of depression following neuronal injury (Moreno-López et al., 2016). Clinical and experimental studies also show that inflammatory cytokines, such as interleukin-6 (IL-6), interleukin-1, and tumoral necrosis factor (TNF-α), are elevated in cerebrospinal fluid and serum shortly after TBI, leading to chronic neuroinflammation (Bodnar et al., 2018; Risbrough et al., 2022; Vedantam et al., 2021). In the central nervous system (CNS), the increased expression of re-uptake transporters triggered by inflammatory processes reduces the metabolism of several neurotransmitters, such as serotonin, dopamine, and norepinephrine. This contributes to neuronal damage and promotes the development of depression after TBI.

Supporting this view, studies using animal models of TBI suggest the involvement of several altered molecular pathways correlated with depressive-like behavior. These alterations include increased phosphorylation of the inhibitory site of glycogen synthase kinase-3 beta in the hippocampus of mice (Shapira et al., 2007), decreased levels of p-extracellular signal-regulated kinase 1/2 and p-p38 mitogen-activated

protein kinase in the hippocampus of rats (Kuo et al., 2013), loss of dopaminergic neurons in the substantia nigra, and reduced dopamine levels in the striatum of rats (Tan et al., 2015). Additionally, systemic changes, such as peripheral inflammation marked by elevated serum levels of peripheral leukocytes, have been observed in male mice (Bahader et al., 2024) (Fig. 1).

Another consequence of the inflammatory process in TBI-induced depression is the activation of the kynurenine pathway (KP), which produces the cellular energy source nicotinamide adenine dinucleotide from tryptophan (TRP) via several neuroactive intermediates (Meier and Savitz, 2022). In both peripheral and CNS, the KP is the primary pathway for TRP metabolism, with 95 % being converted into kynurenine (KYN) by the enzymes indoleamine 2,3-dioxygenase and tryptophan dioxygenase. KYN is then further metabolized into kynurenic acid (KynA) by kynurenine aminotransferases (KATs) or into 3-hydroxykynurenine (3HK) by kynurenine monooxygenase. In the CNS, TRP uptake and metabolism in astrocytes lead to the production of KynA, which is generally considered neuroprotective. In contrast, TRP metabolism in microglia results in the production of neurotoxic metabolites, such as 3HK and quinolinic acid (QuinA), the latter of which acts as an agonist at the *N*-methyl-D-aspartate (NMDA) glutamate receptor and is neurotoxic at high concentrations (Meier and Savitz, 2022) (Fig. 2). In humans, elevated concentrations of 3HK and QuinA, and/or reduced concentrations of KynA, have been associated with depression. Furthermore, these abnormalities in the KP are alleviated by effective treatments for depression, including cognitive behavioral therapy (Savitz, 2020) and physical exercise (Javelle et al., 2021).

After TBI, alterations in neuronal circuits involved in emotional processing, caused by chronic inflammation and elevated levels of QuinA, increase the risk of developing symptoms of depression and anxiety (Morganti-Kossmann et al., 2019). Neural structures like the hippocampus, which has the highest density of NMDA receptors (Monaghan and Cotman, 1985), are particularly vulnerable to QuinA-mediated secondary injury following TBI. Additionally, changes in the metabolism of dopaminergic neurons in regions regulated by glutamatergic and GABAergic afferents, such as the ventral tegmental area (Morales and Margolis, 2017) lead to depressive behavior through KP-mediated changes in NMDA receptor signaling (Bigler, 2007).

5. Molecular effects of physical exercise on TBI-induced depression: An overview

Recent guidelines recommend physical exercise as a therapeutic intervention for individuals with depression (Recchia et al., 2022). Whether used alone or in combination with other strategies, such as antidepressants or psychotherapy, physical exercise can reduce the risk of developing depression (Pearce et al., 2022; Schuch and Stubbs, 2019) and has moderate to large effects on depressive symptoms in clinically diagnosed patients (Heissel et al., 2023). Although the antidepressant effects of physical exercise have been clearly demonstrated, the neurobiological mechanisms underlying these effects remain a topic of debate, with several hypotheses proposed (Ross et al., 2023; Schuch et al., 2016). After TBI, preliminary clinical findings suggest that increasing daily physical exercise helps alleviate symptoms of depression (Bombardier et al., 2017) (see Table 1). (See Table 2.)

Regarding the role of physical exercise in the pathophysiology of depression, several key hypotheses suggest that physical activity affects: 1) oxidative stress, 2) inflammation, 3) neurogenesis and neuronal growth, and 4) structural and functional brain adaptations. It is well known that physical exercise stimulates an acute increase in free radicals in active muscle fibers due to heightened metabolic activity (Powers et al., 2020); however, this transient rise in free radicals is followed by a compensatory response from antioxidant enzymes (Azizbeigi et al., 2014). This effect, known as hormesis, is characterized by a bell-shaped curve response of the antioxidant system, which counteracts the effects of chemicals, toxins, and radiation (Radak et al., 2005). The hormesis hypothesis is supported by animal models and has been observed in healthy or non-depressed populations (Azizbeigi et al., 2014). Notably, a small randomized controlled trial in patients with depression reported a reduction in lipoperoxidation levels following physical exercise intervention (Schuch et al., 2014). In this context, further studies are needed to explore the role of oxidative stress in the antidepressant effects of physical exercise.

Regarding the neurological impairment cause by TBI, a substantial body of evidence suggests that depression resulting from this injury may be, at least in part, due to changes in surviving neurons and dysfunction occurring in the early stages following the injury (Bailes and Mills, 2010). Although it does not directly affect depressive behavior, the acute

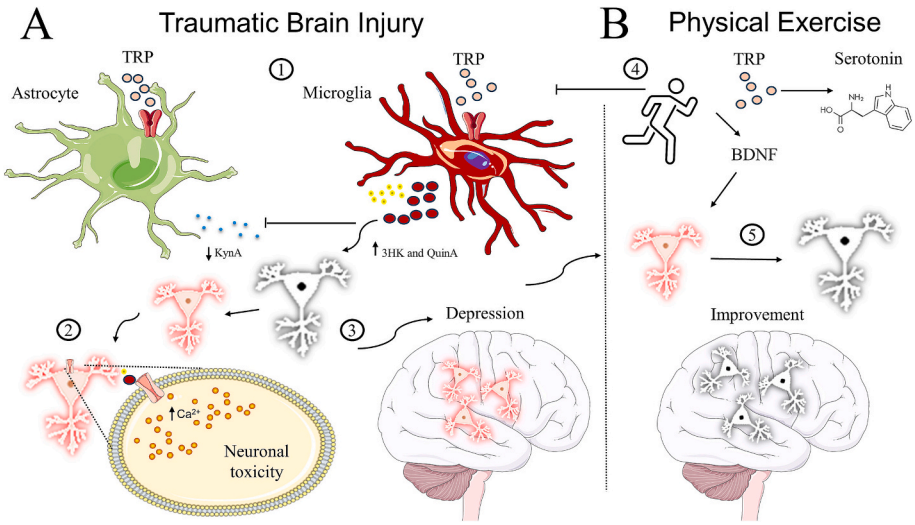

Fig. 2. Proposing Effects of TBI on TRP Uptake and Protection by Physical Exercise. Legend. Mechanisms involved in TBI-induced depression (A) and protective effects of physical exercise (B). 1) After TBI, activated microglia increase tryptophan (TRP) uptake, leading to the production of neurotoxic metabolites such as 3-hydroxykynurenine (3HK) and quinolinic acid (QuinA), which inhibit kynurenic acid (KynA) production by astrocytes. 2) 3HK and QuinA act as agonists at *N*-methyl-D-aspartate (NMDA) receptors, resulting in elevated intracellular calcium concentrations, which contribute to neuronal toxicity. 3) Neuronal toxicity causes cellular loss in critical brain regions involved in mood regulation, leading to depression. 4) Physical exercise can reduce TBI-induced brain inflammation by decreasing microglial TRP uptake and promoting serotonin production, while also increasing brain-derived neurotrophic factor (BDNF) levels. 5) Physical exercise mitigates TBI-induced neuronal loss, promoting neuronal survival in mood-regulating brain areas.

Table 1
Characterization of studies of depression, exercise and traumatic brain injury.

AUTHOR/ YEAR	STUDY DESIGN	SAMPLE PRE/EX			SAMPLE POST/CON			CONTROL GROUP	INTERVENTION	DURATION	FINDINGS
		N	Age	% Female	N	Age	% Female				
Bateman et al., 2001 (Bateman et al., 2001)	RCT		41.7 ± 14.3	44.9		44.7 ± 13.3	31.7	Relaxation training	Cycle ergometer aerobic training	3 months	There were no significant differences for depressive symptoms between the groups.
Bellon et al., 2015 (Bellon et al., 2015)	RCT	69	43.7 ± 15.8	41	69	43.7 ± 15.8	41	Nutrition education module	Home-based walking program	24 weeks	Improvement in depressive symptoms during the program (F = 5.185; $p = 0.007$).
Chin et al., 2014 (Chin et al., 2015b)	Non-controlled trial	7	33.3 ± 7.9	71.4	7	33.3 ± 7.9	71.4	–	Aerobic treadmill exercise	12 weeks, 3 per week	There was no change in depressive symptoms (Δ = −3.1 (4.9); $p = 0.139$).
Damiano et al., 2016 (Damiano et al., 2016)	Non-controlled trial	12	31.3 ± 9.4	41.6	12	31.3 ± 9.4	41.6	Healthy volunteers who were not included in the analyzes	Exercise program using an elliptical trainer	8 weeks, 5 days per week	Improvement in depressive symptoms score ($p = 0.350$).
Driver et al., 2003 (Driver et al., 2003)	Pilot study	3	37	75	3	37	75	–	Aerobic exercises carried out in an aquatic environment, using water bikes	60 min per session, 3 per week for 4 months	Both acute and affective positive increases and increases in physical self-concept and self-esteem
Hoffman et al., 2010 (Hoffman et al., 2010)	RCT	40	39.7	62	40	37.1	50	People with traumatic brain injury	Weekly supervised aerobic exercise sessions	10 weeks, duration of 45 min, 15 of which are warm-ups	Comparisons between groups showed no difference for depressive symptoms ($p < 0.250$).
Rezezak et al., 2015 (Rzezak et al., 2015)	RCT	12	31.8 ± 9.5	0	12	30.5 ± 9.5	0	Paired healthy volunteers	2 cycling sessions	30 min per session	There was no group by condition interaction, but baseline differences between depressed patients and controls (d = 1.18) disappeared after an exercise session.
Schwantd et al., 2012 (Schwandt et al., 2012)	Experimental	4	29	25	4	29	25	–	Exercises with intensity between 5 and 6 on the Borg scale, using cycle ergometer, treadmill and recumbent step machine.	12 weeks, 3 days per week with 10 min of warm-up and 30 min of exercise	Participant scores decreased from moderate to severe and severe levels of depression at baseline to mild to moderate and no symptoms after intervention.
Weisntein et al., 2018 (Weinstein et al., 2017)	Non-controlled trial	10	32.9 ± 6.5	60	10	32.9 ± 6.5	60	–	Aerobic treadmill exercise supervised	12 weeks, 3 per week, 30 min	Improvement in short-term mood responses with reaching 19 % by week 12 ($p = 0.04$, d = 0.74).
Wise et al., 2012 (Wise et al., 2012)	Non-controlled trial	40	39.7 ± 12.6	–	37	39.7 ± 12.6	–	–	Exercise intervention program	Weekly 60 min sessions and 30 min of aerobic exercises 4 times each week	Decrease in BDI depression scores from baseline to 10 weeks ($t = 4.72$; $p = 0.009$)

RCT: Randomized Clinical Trial; BDI: Beck depression inventory; EX: Exercise; CON: Control.

failure of mitochondrial function, followed by inflammatory and oxidative stress, suggests that bioenergetic dysfunction may play a key role in the post-traumatic "window of vulnerability" (Severo et al., 2020). In line with this, it is well known that individuals with mental disorders are often in a systemic pro-inflammatory state (Osimo et al., 2020).

Consistently, people with depression exhibit increased levels of pro-inflammatory interleukins, such as IL-3, IL-6, IL-12, and IL-18 (Osimo et al., 2020). Antidepressants appear to reduce various inflammatory markers, including IL-3, IL-4, IL-7, IL-8, IL-9, IL-13, and IL-17 (Kofod et al., 2022). Similar to oxidative stress, acute physical exercise induces a transient pro-inflammatory state by increasing IL-6, IL-8, and TNF-α (Cerqueira et al., 2020). However, regular physical exercise can lead to adaptations in the inflammatory system, reducing the acute rise in pro-inflammatory cytokines and increasing anti-inflammatory cytokines in non-clinical populations (da Luz Scheffer and Latini, 2020). In individuals with depression, a meta-analysis by Schuch et al. (2016) summarized the available evidence on the effects of exercise on inflammatory markers and found no acute or chronic changes in any marker. It is important to note, however, that some of the included studies had small sample sizes or involved patients undergoing pharmacological treatment (Schuch et al., 2016).

For this reason, animal models offer a valuable opportunity to investigate the neurobiological effects of physical exercise on TBI-induced depression. In experimental models of TBI, reactive activation of microglia (microgliosis) and the release of TNF-α following neuronal

Table 2
Characterization of reviews about depression, exercise and traumatic brain injury.

AUTHOR/ YEAR	TYPE OF REVIEW	INCLUSION CRITERIA	DATA BASE	STUDIES INCLUDED (N)	FINDINGS
Adamson et al., 2015 (Adamson et al., 2015)	Systematic Review and Meta-Analysis	Randomized controlled trials, with physical exercise interventions, including adults, diagnosed neurological disorders, including TBI, with measures of depressive symptoms	CINAHL, Cochrane Register of Controlled Clinical Trials, EMBASE, ERIC, MEDLINE, PsycINFO, PubMed, and SPORTDiscus	26	Overall effect size of 0.28 (95 %CI = 0.15–0.41; $p < 0.001$) favoring a reduction in depression outcomes after an exercise intervention compared with the control condition.
Gertler et al., 2015 (Gertler et al., 2015)	Systematic Review	Randomized controlled trials (RCTs) of non-pharmacological interventions for depression in adults and children who had a TBI.	Cochrane Injuries Group Specialised Register; Cochrane Central Register of Controlled Trials; Database of Abstract of Reviews of Effects; MEDLINE; Embase; CINAHL Plus; PsycINFO; PsycBITE.	6	The results suggest that non-pharmacological interventions, such as physical exercise, may be beneficial in reducing depressive symptoms in this population.
Perry et al., 2020 (Perry et al., 2020)	Systematic Review and Meta-Analysis	Participants ≥18 years old with TBI; Interventions with physical exercise; clinical trials with quantitative data; with a measure of depression-related outcomes; Control group data must be from a comparable population, or the study will be analyzed as an uncontrolled trial.	PubMed, PsychInfo and Web of Science	9	The effect size of exercise for reducing depressive symptoms was small (SMD = 0.48; 95 % CI 0.16 to 0.81).

injury play a role in the pathophysiology of depression (Chiu et al., 2016; Lim et al., 2017). Starting 14 days after TBI, mice subjected to 12 weeks of treadmill exercise (beginning with 5 min at a speed of 8 m/min, followed by 10 min at 12 m/min, and concluding with 5 min at 5 m/min, for a total of 20 min of running per day) showed a reduction in anxiety-like behavior and depression, as well as prevention of aggressive behavior (Bonsale et al., 2023). In the same study, electrophysiological recordings revealed that this physical exercise protocol protected against TBI-induced impairment of long-term potentiation (LTP) (Bonsale et al., 2023), suggesting that physical exercise effectively preserves synaptic plasticity in specific brain circuits that regulate affective behaviors and cognitive tasks. Mice subjected to moderate-intensity treadmill exercise for 44 days also showed a reduction in lesion volume, attenuation of the inflammatory response, and improvements in neurological behaviors (anxiety and depression) induced by TBI through the suppression of the interferon gene stimulatory pathway (STING) (Chen et al., 2023). Given that STING-induced neuroinflammatory responses are known to play a critical role in the detrimental processes following TBI (Hopfner and Hornung, 2020; Zhang et al., 2022), it is plausible to propose that adaptations in the inflammatory system induced by physical exercise protect against TBI-induced psychiatric disorders.

Another important substance released by brain cells is BDNF, a widely studied neurotrophic compound responsible for reorganizing synaptic plasticity during the "reconnection" phase of brain circuitry after TBI (Royes and Gomez-Pinilla, 2019). Although the reduction in BDNF is not specific to depression (Jing et al., 2019), clinical data reveal a relationship between decreased serum BDNF levels and the severity of the injury (Kalish and Phillips, 2010). In experimental models of TBI, a decrease in hippocampal BDNF is associated with depression and spatial memory impairment (Chen et al., 2005). Experimental data also indicate that mild TBI during adolescence, even when it does not cause motor deficits or histological damage, can reduce hippocampal BDNF expression, leading to long-term depressive behavior in adulthood (Yang et al., 2021). Furthermore, BDNF is a moderator and predictor of treatment response, with higher baseline BDNF levels associated with better outcomes following antidepressant treatment (Carniel and da Rocha, 2021). There is some evidence suggesting that higher BDNF levels are linked to a greater response to physical exercise (Ribeiro et al., 2021). However, limited evidence exists regarding the role of BDNF levels in individuals with depression (Dinoff et al., 2018; Kurebayashi and Otaki, 2018). While this hypothesis cannot be fully accepted, it should not be dismissed, as existing trials have significant limitations, such as the concurrent use of pharmacological antidepressants. After TBI, it is plausible that variations in serum levels of this neurotrophic factor may reflect the severity of post-TBI depression, suggesting a potential biomarker to track the response and effectiveness of physical exercise treatment.

Lastly, brain imaging studies have revealed several abnormal changes in brain structure and function in people with depression (Han et al., 2023). There is evidence of decreased gray matter volume in the frontal, temporal, and limbic regions (Jiang et al., 2023). Regarding connectivity, studies have demonstrated hyper-connectivity between the default mode and frontoparietal networks, as well as hypo-connectivity between the limbic and salience networks in individuals with depression (Brandl et al., 2022). Clinical data from sixteen participants with mild to severe TBI and persistent self-reported neurological symptoms revealed that cerebral blood flow deficits, particularly in the hippocampus and rostral anterior cingulate, were associated with higher levels of self-reported symptoms of anger, anxiety, and depression. This suggests that lower cerebral blood flow in the hippocampus and rostral anterior cingulate may play a role in self-reported affective symptoms (Thomas et al., 2021).

In the sporting environment, high-profile athlete suicides and behavioral disturbances have drawn public attention to the link between repeated concussive injuries and poor psychosocial outcomes later in life (Kuhn et al., 2017; Rice et al., 2018). Although macrostructural and morphological changes are typically absent in the acute phase of sport-related concussion, clinical findings have reported decreases in total and regional brain volume, as well as cortical thickness, in athletes months and years after the injury (Mayer et al., 2023; Urban, 2021). On the other hand, individuals with TBI who underwent 12 weeks of supervised moderate-intensity aerobic cycling showed greater increases in the volume of the left hippocampus and right thalamus, along with improvements in memory and processing speed (Wender et al., 2021). After TBI, numerous studies in both rodents and humans indicate that exercise improves brain function; however, significant gaps and limitations in the current research landscape need to be addressed. For instance, while the positive impact of physical exercise has been consistently demonstrated in healthy older adults (Kramer et al., 2003), other studies have failed to replicate this effect in individuals with TBI (Grealy et al., 1999). Although there is a consensus on the safety and effectiveness of aerobic exercise as a post-TBI rehabilitation strategy, future studies are needed to determine how structural brain changes

resulting from physical exercise may serve as a protective mechanism against TBI-induced psychiatric disorders.

6. Clinical deficits and molecular mechanisms of central fatigue

Fatigue is characterized by feelings of physical and/or mental tiredness, weakness, lack of energy, slowness, and difficulty concentrating (Finsterer and Mahjoub, 2014; Penner and Paul, 2017). It is a common manifestation of various physical, psychological, and neuropsychiatric disorders (Abd-Elfattah et al., 2015). Fatigue can make it challenging to initiate or maintain voluntary and physical activities (Chaudhuri and Behan, 2004). Given the complexity of the biological and behavioral phenomena involved, a comprehensive definition of fatigue remains a topic of ongoing discussion (Aaronson et al., 1999; Billones et al., 2021; Penner and Paul, 2017).

Fatigue is a heterogeneous, multifactorial condition (Campos et al., 2022; Landmark-Høyvik et al., 2010), that is objectively manifested through fatigability observed in tasks involving physical or cognitive performance (Finsterer and Mahjoub, 2014; Kluger et al., 2013; Penner and Paul, 2017). Perceptions of fatigue and fatigability may be independent; however, the relationship between these two phenomena is crucial. In healthy individuals, fatigue is a physiological, predictable, and transient response to intense and prolonged activities (Finsterer and Mahjoub, 2014; Kluger et al., 2013). In contrast, individuals with illnesses may experience fatigue even at rest, which can be disabling and limit their usual activities, negatively impacting their physical, social, occupational, and emotional functioning (Finsterer and Mahjoub, 2014).

Fatigue can generally be classified as acute or transient (such as after exercise) (Walters et al., 2019), persistent (lasting one month or more), or chronic (persisting or recurrent for six consecutive months or more) (Fukuda et al., 1994). Its classification is divided into (i) physical fatigue (central and peripheral) related to movements and (ii) mental fatigue related to cognitive tasks (Ishii et al., 2014; Karshikoff et al., 2017). Central fatigue is associated with the supraspinal and spinal regions of the nervous system (Finsterer and Mahjoub, 2014) which reduce the excitation of motor neurons (Boyas and Guével, 2011). Peripheral fatigue relates to the neuromuscular junction and changes in muscle tissue (Boyas and Guével, 2011; Chaudhuri and Behan, 2004) and can also be influenced by mental fatigue (Marcora et al., 2009; Proost et al., 2022). Additionally, psychological fatigue—distinct from mental fatigue—is associated with stress, anxiety, and depression (Walters et al., 2019).

7. Fatigue assessment

Fatigue can be assessed as a symptom (subjective fatigue) or through fatigability (objective fatigue) (Campos et al., 2022; Kluger et al., 2013). Structured questionnaires are widely used due to their low cost and applicability. Most of these instruments are generic, though some are validated for patients with neurological diseases, albeit with limitations in their psychometric properties (Tyson and Brown, 2014). One-dimensional questionnaires focus on assessing symptom severity, while multidimensional instruments evaluate several domains of fatigue, making them more comprehensive in their construct (Whitehead, 2009).

Interestingly, fatigue is assessed not only as a symptom, but also through complementary measures of fatigability. Fatigability can be quantified through physical and/or cognitive performance tests (Campos et al., 2022; Finsterer and Mahjoub, 2014) including physical performance tests and executive function assessments (Campos et al., 2022). Additionally, techniques such as transcranial magnetic stimulation, electroencephalograms, and imaging tests like magnetic resonance imaging and computed tomography have also been employed to evaluate fatigue in neurological disorders (Chaudhuri and Behan, 2004; Lou, 2012).

In this context, the choice of fatigue assessment method should be based on the pathophysiology of the underlying neurological disease, recognizing that fatigue and fatigability are independent phenomena. Furthermore, evaluating associated symptoms—such as anxiety, depression, sleep disorders, pain, and stress—is essential for accurate diagnosis and necessitates a complementary approach for individuals experiencing fatigue.

8. The mechanism of fatigue after TBI

Although clinical and epidemiological knowledge about fatigue in TBI has advanced (137, 138), its etiology remains poorly understood. The difficulty in reliably identifying post-TBI clinical fatigue stems from its subjective nature. Unlike "peripheral fatigue," which is characterized as fatigue in a specific group of muscles after a period of exertion, post-TBI fatigue is defined by central fatigue resulting from neuronal damage (Englander et al., 2010a).

Clinical studies report that central fatigue can occur weeks and months after injury in patients discharged from emergency departments following concussive injuries (Cantor et al., 2013). When these symptoms persist, they can significantly alter quality of life, leading to deterioration in general health, an increased risk of depression, social isolation, and impaired work capacity (Hawthorne et al., 2009).

A theoretical explanation proposed by Ronnback et al. (2004) suggests that inflammatory processes contribute to a decrease in astroglial clearance of extracellular glutamate and the disintegration of the blood-brain barrier following TBI (Hansson and Rönnbäck, 2004). Chronic inflammation after TBI, particularly in subcortical regions with increased microglial activation, also compromises neuronal supply, resulting in reduced glutamate transmission. This hypothesis is based on the short- and long-term inflammatory processes generated by microglial activation that occurs distant from the initial injury site, as observed in experimental models of TBI (da Silva Fiorin et al., 2024b; da Silva Fiorin et al., 2020; Loane et al., 2014).

Clinical studies of TBI-induced mental fatigue indicate that dysfunctions in the cortico-striatal-thalamic circuits, which regulate emotional behavior, are involved in the pathophysiology of this neuronal injury (Johansson et al., 2015; Kohl et al., 2009). After TBI, the chronic downregulation of norepinephrine receptors, coupled with decreased dopamine signaling in experimental models (McGuire et al., 2019), suggests that energy-deprived neuronal networks alter enhanced dopamine signaling and norepinephrine, leading to mental fatigue after neuronal injury.

Using functional magnetic resonance imaging, clinical studies reveal that the subjective sensation of cognitive fatigue (Kohl et al., 2009) correlates with increased physiological activity in areas such as the thalamus and caudate nucleus (Wylie et al., 2023). In the sports environment, the long-term neurobehavioral sequelae (depression and fatigue) caused by repeated mild TBI are related to hypopituitarism and, consequently, GH deficiency. The GH deficiency observed in experimental models of TBI reinforces the involvement of systemic inflammation and persistent astrocytosis in the hypothalamus/pituitary axis leading to hypopituitarism (Kasturi and Stein, 2009) (Fig. 3).

In this context, detecting hypopituitarism in patients with TBI and monitoring treatment effects—such as changes in cognitive functioning and mental distress with hormone replacement therapy—presents challenges for endocrinologists (Pavlovic et al., 2019). Understanding the mechanisms by which the brain interacts with systemic physiology throughout the course of TBI pathophysiology is essential for establishing scientifically based rehabilitative strategies. For instance, increases in frontal cortical thickness, gray matter volume, and resting-state connectivity changes in regions associated with somatosensory networks in TBI patients receiving GH replacement have been associated with significant improvements in self-rated fatigue (Wright et al., 2020). Furthermore, increases in resting energy expenditure, improvements in body composition, and changes in the perception of submaximal effort observed in TBI patients supplemented with GH and subjected to

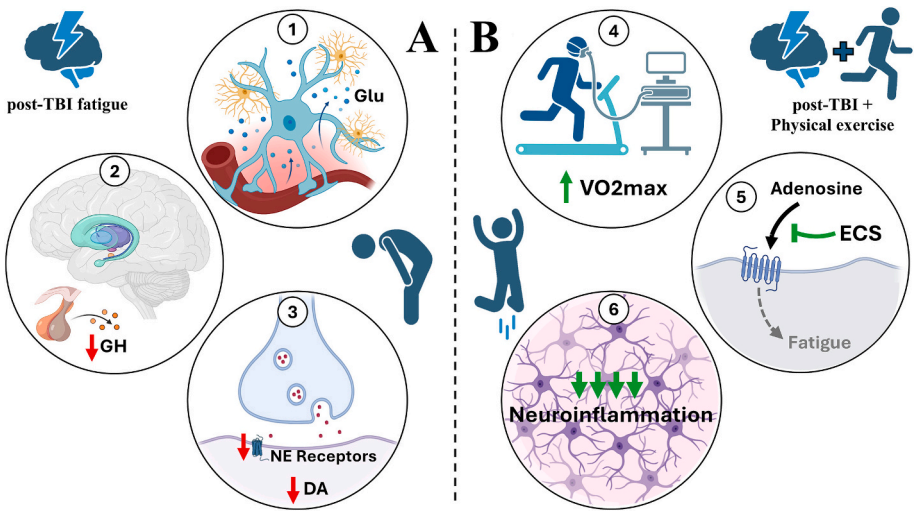

Fig. 3. Effects of physical exercise on post-TBI Fatigue. Legend. Although the cause of fatigue after TBI is not fully understood, there are several mechanisms that seem to be related (A), and physical exercise appears to influence some of these mechanisms (B). 1) Post-TBI fatigue is linked to neuroinflammation, which includes reduced astroglial clearance of extracellular glutamate (Glu), disruption of the blood-brain barrier, and effects on astroglial metabolism and function. These factors may impair microglial support to neurons, leading to diminished glutamate transmission. 2) Dysfunctions in the cortico-striatum-thalamic circuits following TBI, such as increased physiological activity, regulate emotional behavior and are strongly correlated with the subjective experience of cognitive fatigue. Additionally, systemic inflammation and persistent astrocytosis in the hypothalamus and anterior pituitary may contribute to hypopituitarism after TBI, linking reduced Growth hormone (GH) production to mental fatigue. 3) The chronic downregulation of norepinephrine (NE) receptors that is associated with decreased signaling of dopamine (DA) are found in experimental models of TBI also appear as one of the related mechanisms. 4) On the other hand, Improvements in cardiorespiratory fitness, as measured by maximal oxygen consumption (VO_2 max), may help alleviate depressive symptoms and fatigue following neuronal injury. 5) In addition, endogenous substances elevated during physical exercise, such as endocannabinoids (ECS), inhibit the uptake of adenosine in the striatum, a factor highly related to central fatigue. 6) Moreover, classical general benefits from physical exercise, such as increase in resting energy expenditure, the improvement in body composition and inflammatory control, seems to act in all the mechanisms cited before, improving fatigue sensation.

progressive exercise testing (Wright et al., 2024) suggest the involvement of the GH pathway in physical and mental improvement after TBI.

Notably, experimental and clinical studies report physiological alterations in sympathetic and parasympathetic tone after TBI, characterized by an increased heart rate (do Nascimento et al., 2023; Meyfroidt et al., 2017), which may contribute to chronic fatigue. In fact, the autonomic nervous system (ANS) is known to be altered in individuals with chronic fatigue syndrome (Van Cauwenbergh et al., 2014). Moreover, decreased parasympathetic activity and increased sympathetic activity are associated with CNS fatigue (Tanaka et al., 2009), suggesting that ANS dysfunction following TBI may drive the central fatigue process.

9. Effects of physical exercise on post-TBI fatigue

Regarding the pathophysiological mechanisms associated with TBI and fatigue, physical exercise has been shown to improve mood and reduce fatigue symptoms (Driver and Ede, 2009). Although the exact molecular signaling pathways have not yet been fully defined, it has been suggested that the benefits of physical exercise on fatigue include an increase in maximal oxygen consumption (VO_2max), which is 25 % lower in individuals with TBI (Mossberg et al., 2007). The cardiorespiratory improvements, characterized by increases in VO_2 peak, work rate, and time to voluntary exhaustion in patients with TBI undergoing aerobic training (Chin et al., 2015a) suggest that enhanced cardiorespiratory fitness may alleviate depressive symptoms and fatigue following neuronal injury (Driver and Ede, 2009).

To describe the level of physical activity in adolescents and young adults after TBI, as well as its association with injury characteristics and fatigue symptoms, clinical studies reveal that a lack of motivation is related to physical inactivity (Driver et al., 2012). This underscores the need to address both the physical and mental aspects of physical activity in TBI patients. Sedentary behaviors, such as watching television or using computers, tablets, or smartphones, observed in adolescents with TBI (van Markus-Doornbosch et al., 2017) further reinforce this understanding.

It is important to consider, however, that certain details of physical exercise protocols, such as duration and intensity, may have contradictory effects on CNS function. For instance, experimental studies have demonstrated that increased adenosine in the striatum alters cortico-striatal neuroplasticity in exercise-induced fatigue (Aguiar Jr et al., 2020). Strenuous physical exercise also elevates endogenous substances like endocannabinoids, which inhibit the uptake of adenosine in the striatum (Pandolfo et al., 2011), a metabolic factor involved in central fatigue (Ma et al., 2018).

On the other hand, elevated levels of anandamide in individuals engaging in running or cycling at 70 % to 80 % of their maximum heart rate for 45 min suggest that endocannabinoids act on peripheral sensory fibers to reduce pain (Sparling et al., 2003). After TBI, low brain endocannabinoid levels, associated with persistent, undirected nocturnal hyperactivity and central fatigue, suggest that deficiencies in perilesional and subcortical regions contribute to post-TBI psychopathology (Vogel et al., 2020).

Considering that the endocannabinoid system is an extensive endogenous signaling system with multiple elements and its activation reduces stress and anxiety while improving well-being (Patel et al., 2017), it is plausible to propose that procannabinoid-directed therapeutic or preventive approaches, such as physical exercise, may improve post-TBI psychopathology (Fig. 3). In particular, the protective effects of physical exercise against the persistent loss of endocannabinoids in the cortex and subcortex may represent a promising opportunity for the development of a new class of therapeutic applications.

10. Sleep restriction and deprivation

Insufficient sleep syndrome, defined as hypersomnia symptoms caused by sleep restriction (Sateia, 2014), can affect individuals of both sexes and all ages, compromising their perceptions and sensations

(Kundermann et al., 2004). Lack of adequate sleep impacts emotions (Killgore et al., 2017) and decision-making (Killgore, 2010). These effects of insufficient sleep offer insights into how this behavior can negatively influence relationships and human resilience, potentially impacting mental health (Chattu et al., 2019).

Genetic vulnerabilities, such as specific gene mutations, can induce changes in biological rhythms and sleep requirements (Bringmann, 2019; Bunney and Bunney, 2000). These circadian disturbances can alter the rhythmic activity of neurotransmitters involved in mood regulation, such as dopamine and serotonin (McClung, 2007), leading to the development of depressive symptoms (Daut and Fonken, 2019). Circadian abnormalities, particularly delayed rhythms, may play a role in the pathogenesis of depressive disorders (Bunney and Bunney, 2000).

Considerable evidence suggests that neuropsychiatric diseases are associated with changes in circadian rhythms (Jagannath et al., 2013; Wulff et al., 2010). Notably, fibroblasts isolated from patients with schizophrenia show a loss of key elements, including circadian regulators such as period (PER) and cryptochrome (CRY) expression. Additionally, their peripheral blood leukocytes exhibit decreased and/or disrupted diurnal expression of core circadian clock genes—CLOCK, PER1/2/3, CRY1—and a functional homologue of CLOCK, neuronal PAS domain protein 2 (NPAS2), compared to healthy control subjects (Johansson et al., 2016). Fibroblasts isolated from patients with bipolar disorder show greater variation in period and amplitude, along with deficits in entrainment pathways. Consequently, lithium is used to treat manic-depressive disorders, with its main therapeutic target postulated to be the nuclear receptor Rev-erbα (Yin et al., 2006). Patients with major depressive disorder also exhibit significant disruptions in circadian rhythmicity and phase shifts of central clock genes in several regions of the brain (Li et al., 2013).

The metabolic system also exhibits circadian control, influencing the cycling of multiple metabolic pathways in the liver (Kornmann et al., 2007). Processes such as glucose, cholesterol, and triglyceride metabolism are examples, with the rate-limiting steps in these processes being key sites of circadian regulation. Rhythmic disturbances in pancreatic islets lead to broad transcriptome variations, affecting the expression of genes involved in cell survival, growth, and synaptic vesicle formation. Polymorphisms in the CLOCK gene and basic helix-loop-helix ARNT-like 1 (BMAL1) have been associated with metabolic disorders and linked to increased susceptibility to obesity (Scott et al., 2008; Sookoian et al., 2008). Additionally, two BMAL1 haplotypes have been associated with hypertension and type 2 diabetes mellitus (Woon et al., 2007).

Similar studies have linked polymorphisms in other core clock genes, such as PER2 and NPAS2, to fasting hyperglycemia and hypertension, respectively (Englund et al., 2009). Sleep deprivation has also been shown to alter the plasma metabolome (Davies et al., 2014) and urinary metabolome (Giskeødegård et al., 2015). Reduced sleep time is associated with significant changes in the lipid metabolome (Depner et al., 2020). Weljie and colleagues identified changes in the lipid metabolome—specifically lysophosphatidylcholines, triglycerides, phosphatidylcholines, and acylcarnitines—in response to sleep restriction (limited to 4 h) and found that one night of sleep was insufficient to restore the decreased levels of long-chain acylcarnitines (Weljie et al., 2015). Insufficient sleep also affects metabolites related to mitochondrial fatty acid oxidation, resulting in increased levels of acylcarnitines in both healthy individuals and patients with diabetes (van den Berg et al., 2016).

The detrimental effects of sleep restriction on metabolic health are also supported by metabolomic studies, which have observed a decrease in high-density lipoproteins in groups experiencing prolonged sleep restriction (Aho et al., 2016). Sleep fragmentation, common in modern society, has been linked to a reduction in hippocampal volume, deficits in learning and cognition, and decreased neurogenesis (Navarro-Sanchis et al., 2017). Overall, these results suggest that branched-chain amino acid metabolism is associated with sleep-wake regulation. The three essential branched-chain amino acids—L-leucine, L-valine, and L-isoleucine—must be obtained through diet, as they are not produced by the body (Wolfe, 2017).

In summary, sleep deprivation has both global and specific effects on cognition, moral judgment, and behavioral responses, which are associated with altered functioning of the prefrontal cortex and parietal regions. Sleep deprivation also disrupts normal affective processing, likely by weakening prefrontal inhibitory systems, allowing emotional systems to function relatively unchecked, without proper modulation or integration. Additionally, glucose and lipid metabolism, as well as antioxidant status, are affected by changes in biological rhythms. Other studies conducted on more relevant sampling substrates to identify biomarkers of mental disorders have relied on animal models, as brain-derived samples from humans are typically only available postmortem.

11. Sleep disorders following TBI

Associated with persistent neuropsychiatric symptoms such as depression, anxiety, and fatigue, sleep disturbances are among the most common problems following TBI, impeding recovery for many years after the injury (Bell et al., 2018). Although sleep disturbances and fatigue can occur independently, they often happen simultaneously, with one worsening the other, creating a vicious cycle. It is important to note that sleep disorders are more common among individuals with TBI compared to the general population. Clinical studies in rehabilitation units reveal that up to 68 % of patients experience sleep disorders (Makley et al., 2008). Approximately 46 % of patients with subacute TBI have abnormal sleep patterns, and 25 % experience excessive daytime sleepiness (Castriotta et al., 2007). Chronically, more than a quarter of TBI patients continue to suffer from significant sleep disturbances (Colantonio et al., 2010), including insomnia and circadian rhythm disruptions (Barshikar and Bell, 2017; Nakase-Richardson et al., 2013). However, the mechanisms underlying this bidirectional relationship remain largely unknown.

Clinical studies demonstrate that structural brain lesions, including damage to the suprachiasmatic nucleus, hypothalamus, and amygdala in patients with severe TBI, contribute to secondary sleep-wake disturbances. Patients with mild TBI, on the other hand, often do not have abnormal radiographic findings, and direct pathological examination is rarely possible (Hou et al., 2013; Yassin et al., 2015). This lack of a clear conceptual framework complicates both measurement approaches and potential treatments.

Given the impact of sleep disturbances on the brain after TBI, it has been suggested that sleep disruption may affect recovery following neuronal injury. Like in humans, experimental TBI models indicate that hypersomnolence, characterized by slow oscillatory activity in the delta frequency range, is crucial for facilitating cognitive improvement after neuronal injury (Morawska et al., 2016). Interestingly, sleep induction with the drug gamma-hydroxybutyric acid and/or sleep restriction enhances encephalographic slow-wave activity, significantly reduces diffuse axonal damage in the cortex and hippocampus, and prevents the development of cognitive impairment (Morawska et al., 2016). Although these findings may seem contradictory, they are attributed to an increase in delta power (slow-wave sleep) (Martinez-Vargas et al., 2012; Mulkey et al., 2001).

The mechanism underlying delta power-associated improvements in neurocognition has been linked to the recently described cerebral glymphatic system, which shows increased clearance of toxins and waste from the cerebral interstitium during sleep (Sullan et al., 2018). After TBI, inflammation, accumulation of acidic protein residues (e.g., cleaved tau, phosphorylated tau [p-tau], and amyloid beta [Aβ]), and astrocytic proteins (e.g., glial fibrillary acidic protein and S100 calcium-binding protein B) are implicated in glymphatic system dysfunction (Jessen et al., 2015). The reduced clearance potential of this system leads to an accumulation of neurotoxins, such as Aβ and tau byproducts, in extracellular spaces (Iliff et al., 2014; Ren et al., 2017), contributing to

the progression of neurodegenerative processes like Alzheimer's disease (Tarasoff-Conway et al., 2015) and chronic traumatic encephalopathy (Kanaan et al., 2015).

Given that sleep is essential for the optimal functioning of the glymphatic system, it is plausible that chronic sleep disruption may exacerbate glymphatic dysfunction, increasing the accumulation of Aβ and tau products after TBI. Furthermore, the abnormal expression of circadian genes such as CLOCK, PER2, and BMAL1 following neurological injury provides a theoretical basis for treating sleep disorders by regulating biological rhythms (Boone et al., 2012; Zhanfeng et al., 2019). After TBI, disruptions in signaling from the suprachiasmatic nucleus of the hypothalamus, followed by disturbed rhythms in individual peripheral structures caused by the immune response, also alter clock gene expression and lead to circadian desynchronization. Additionally, positive feedback in peripheral structures increases cytokine-induced inflammation, which can cross the blood-brain barrier and further exacerbate desynchrony. This loss of phase coherence between rhythmic structures may significantly contribute to negative symptomatology after TBI (Korthas et al., 2022; Yamakawa et al., 2020). However, circadian desynchronization caused by the immune response following TBI remains understudied, and future research is needed to clarify this mechanism.

12. Sleep disturbances after TBI – therapeutic approach from physical exercise

It is widely accepted that patients with TBI report sleep-wake disturbances, such as excessive daytime sleepiness and insomnia. In this context, excessive daytime sleepiness has been shown to correlate with lower hypocretin-1 levels in the cerebrospinal fluid of TBI patients six months after injury (Baumann et al., 2007). Considering that hypocretin-1 is a neuropeptide exclusively produced by a group of neurons in the lateral and dorsomedial hypothalamus, which project throughout the brain (Fronczek et al., 2008), it is plausible to suggest that a long-term downregulation of wake-promoting neurotransmitters following TBI affects sleep (Fig. 4).

It is also important to note that the combination of increased inhibition of excitation-related nuclei projecting to the ventrolateral preoptic area, along with a heightened sensitivity of orexin neurons to GABA in response to sleep deprivation, reduces sensitivity to endogenous stimuli that enhance energy expenditure. This, in turn, can stimulate weight gain (DePorter et al., 2017). This hypothesis is supported by the prevalence of sleep disorders diagnosed in TBI patients, which contribute to symptoms commonly seen in a sedentary lifestyle, such as obesity, diabetes, and congestive heart failure (Folmer et al., 2020). In the sports environment, systematic and meta-analytic studies demonstrate that sleep disturbances impair recovery, whereas early physical activity (aerobic exercise) following neuronal injury can reduce recovery delays and sleep disturbances (Leddy et al., 2023; Sawyer et al., 2016).

Regarding the role of physical exercise in TBI-induced sleep disorder (Lowe et al., 2020), interventions involving exercise programs in adults with a history of TBI have shown that physical exercise (15 min of education, 15 min of warm-up activities, and 30 min of aerobic exercise at least four times a week) for 10 weeks can protect against TBI-induced sleep disturbances (Pittsburgh Sleep Quality Index [PSQI] > 5) (Hoffman et al., 2010). Conversely, TBI patients who participated in a supervised program of vigorous treadmill exercise (30 min, three times a week) showed improvements in cognitive function but not in sleep function immediately after the intervention (Chin et al., 2015b). These conflicting findings highlight the need to consider the distinct neuroplasticity processes elicited by different exercise protocols following TBI.

It is well known that sleep and physical activity are modifiable behavioral factors associated with better health. However, after TBI, the bidirectional relationships between sleep and physical activity are still in the early stages of understanding. In this context, clinical data from 64 individuals with TBI who reported sleep problems or fatigue showed no significant association between daytime physical activity and sleep duration or continuity. Additionally, less sleep time and reduced wakefulness after sleep onset were associated with more time spent in light-intensity activities, suggesting that sleep may influence physical activity more than physical activity influences sleep in patients with TBI (Bruijel et al., 2021).

Although the beneficial effects of physical activity on sleep are not completely understood, improving sleep quality may be relevant to

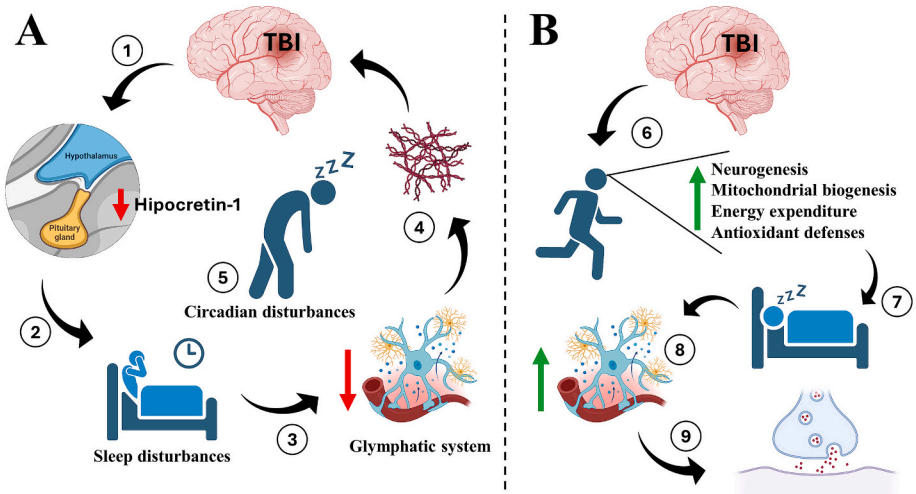

Fig. 4. Effects of physical exercise on post-TBI sleep disturbances. Legend. Sleep disturbances are one of the most common problems following TBI (A). 1) Structural brain lesions, including those in the suprachiasmatic nucleus of the hypothalamus, may be related to a low concentration of hypocretin-1 in cerebrospinal fluid; 2) a neuropeptide highly associated with sleep disturbances. 3) Poor sleep quality prevents the glymphatic system from clearing toxins and residues produced during the day as well as the secondary damage from traumatic brain injury. 4, 5) This accumulation of residues, such as p-tau tangles and β-amyloid plaques, may exacerbate the TBI damage and perpetuate the cycle of circadian disturbances. On the other hand, the use of physical exercise as a non-pharmacological approach seems to improve sleep quality (B). 6, 7) Although the exact mechanisms are still unclear, the increased neurogenesis, mitochondrial biogenesis, antioxidant defenses, and energy expenditure induced by physical exercise may be behind the improvement in sleep. 8, 9) This, in turn, supports the proper function of the glymphatic system, has a direct impact on neurotransmitter production, and helps correct circadian rhythm.

supporting a physically active lifestyle in people with TBI. In experimental models of mood disorders, sleep deprivation causes cognitive deficits, while treadmill running training promotes LTP in the CA1 area of the hippocampus, a neurological event associated with improvements in memory tasks (Zagaar et al., 2012). Physical exercise also induces BDNF and phosphorylated calcium/calmodulin-dependent protein kinase II, proteins linked to cognitive improvements (Zagaar et al., 2012).

It is important to note that a lack of BDNF alone may not be sufficient to explain the pathology of stress-related sleep disorders (Autry and Monteggia, 2012); however, it certainly plays a prominent role in the pathophysiology of stress-associated mood disorders (Schmitt et al., 2016). In line with this view, the effects of physical exercise on increasing antioxidant defenses (Marques-Aleixo et al., 2012), improving mitochondrial biogenesis (Navarro et al., 2004), and upregulating the neurotrophin BDNF (Gomez-Pinilla and Hillman, 2013) vividly demonstrate exercise's capacity to promote metabolic homeostasis after TBI (Fig. 4). Consequently, it seems likely that effective control of cellular metabolism, plasma membrane integrity, and synaptic plasticity (Royes and Gomez-Pinilla, 2019) induced by physical exercise may protect against TBI-induced sleep dysfunction. However, there is limited or no evidence to suggest that exercise modifies BDNF levels in individuals with sleep disturbances following neuronal injury, and future studies are necessary to clarify this point.

13. Concluding Remarks

An increasing body of evidence reveals that the pervasive effects of TBI on psychiatric disorders involve a complex interplay of biological alterations in the neural metabolism of structures crucial for social behavior. These multiple pathophysiological processes, at least in part, complicate our understanding of how the brain's inability to maintain metabolic homeostasis may contribute to the worsening of symptoms in TBI-induced psychiatric disorders. On the other hand, there is a consensus that changes in the architecture and neural metabolism of structures involved in the psychiatric behavior of individuals affected by TBI are central to the interface of this behavioral disorder. Understanding how neuronal injury impacts emotional behavior is essential for planning effective treatments to address the long-term burden of living with TBI.

In this context, the present review provides important information about physical exercise as a strategy to limit the TBI-induced pathophysiology characterized by secondary neuronal death, neural repair, and behavioral disorders. It is important to note, however, that situations arising from inappropriate post-injury time windows and different exercise protocols may alter this concept and exacerbate TBI physiopathology. Given the extent of post-traumatic changes in neuronal function and the potential for amplifying secondary cascades that lead to psychiatric disorders, a better understanding of the mechanisms involved in the protective effects elicited by this non-pharmacological intervention will facilitate the successful use of exercise as a therapeutic tool.

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Declaration of competing interest

There are no conflicts of interest to report for any of the authors.
TBI: Traumatic brain injury;

Data availability

No data was used for the research described in the article.

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